

Douglas Rachford Updates

$$\min: f(x) + g(x), \quad \boxed{0 \in \partial f(x) + \partial g(x)}$$

Fixed points of $C_g \circ C_f$ solve $\uparrow$

DR: $\frac{1}{2} (I + C_g \circ C_f)$ firmly non-expansive

$$\begin{cases} x_{k+1} = \text{prox}_f(y_k) \\ y_{k+1} = y_k + \text{prox}_g(2x_{k+1} - y_k) - x_{k+1} \end{cases}$$

$$\begin{cases} u_{k+1} = \text{prox}_g(x_k + w_k) \\ x_{k+1} = \text{prox}_f(u_{k+1} - w_k) \\ w_{k+1} = w_k + x_{k+1} - u_{k+1} \end{cases} \quad \left. \vphantom{\begin{cases} u_{k+1} = \text{prox}_g(x_k + w_k) \\ x_{k+1} = \text{prox}_f(u_{k+1} - w_k) \\ w_{k+1} = w_k + x_{k+1} - u_{k+1} \end{cases}} \right\} \text{Need } \text{prox}_g; \text{prox}_f \text{ eval}$$

Applying to: $\min: \lambda f(x) + \lambda g(x), \quad \lambda > 0$

have $\text{prox}_{\lambda g} \stackrel{i}{=} \text{prox}_{\lambda f}$.

Sparse Inverse Covariance Selection

$$\min_X \underbrace{\langle C, X \rangle - \log \det X}_f + \underbrace{\gamma \sum_{i,j} |X_{ij}|}_g$$

Exercise :

$$\left. \begin{aligned} u_{k+1} &= \text{prox}_{\lambda g}(x_k + w_k) \\ x_{k+1} &= \text{prox}_{\lambda f}(u_{k+1} - w_k) \\ w_{k+1} &= w_k + x_{k+1} - u_{k+1} \end{aligned} \right\} \text{write these updates explicitly}$$

$\text{prox}_{\lambda g}$: soft-thresholding

$\text{prox}_{\lambda f}$: $\log \det X \rightarrow X^{-1}$

$$\hat{X} = \text{prox}_{\lambda f}(Z) : 0 \in C - \hat{X}^{-1} + \frac{1}{\lambda} (\hat{X} - Z)$$

$\hat{X}, \hat{X}^{-1}$ have same eigenvectors. $\rightarrow (\hat{X} - \lambda \hat{X}^{-1}) = (Z - \lambda C) = \sum \mu_i v_i v_i^T$